Han Zhou

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Employment

- **University of Pennsylvania** Philadelphia, PA, USA
  AMCS/CMB Postdoctoral Fellow, Department of Mathematics *Sep. 2024 -*

Education

- **Shanghai Jiao Tong University** Shanghai, PRC
  Ph.D., Mathematics, Advisor: Wenjun Ying. *Sept. 2020 - Jun. 2024*
- **Shanghai Jiao Tong University** Shanghai, PRC
  B.S., Mathematics and Applied Mathematics *Sept. 2016 - Jun. 2020*

Research Interest

- **Scientific Computing & Numerical Analysis**: efficient numerical methods for interface problems and fluid-structure interaction
- **Mathematical Biology**: modeling and simulation of biological membranes, cells, and vesicles
- **PDE Analysis**: well-posedness for fluid-structure interaction and open-boundary systems

Papers and Preprints

1. Sho Kawakami, **Han Zhou**, Po-Chun Kuo, Yoichiro Mori, Yuan-Nan Young. Stability and equilibria of a compressible elastic membrane in Stokes flow. arXiv:2607.03357.
2. Shuo Ling, Wenjun Ying, **Han Zhou**. Principal-part decomposition for neural operator learning of Dirichlet-to-Neumann maps. arXiv:2606.25952.
3. Charles L. Epstein, Yoichiro Mori, **Han Zhou**. Bulk-surface coupled PDE with an open boundary. arXiv:2604.20798.
4. **Han Zhou**, Wenjun Ying. A correction function-based KFBI method for Brinkman interface problems. arXiv:2604.15509.
5. **Han Zhou**, Yuan-Nan Young, Yoichiro Mori. Modeling and simulation of open membranes in Stokes flow with mixed-dimensional coupling. *Multiscale Modeling & Simulation*, 24(2):474–500, 2026. DOI.
6. Pengsong Yin, Wenjun Ying, Yulin Zhang, **Han Zhou**. A kernel-free boundary integral method for elliptic interface problems on surfaces. *Journal of Computational Physics*, 562:115004, 2026. DOI.
7. **Han Zhou**, Shuwang Li, Wenjun Ying. An alternating direction implicit method for mean curvature flows. *Journal of Scientific Computing*, 101:65, 2024. DOI.
8. **Han Zhou**, Wenjun Ying. A correction function-based kernel-free boundary integral method for elliptic PDEs with implicitly defined interfaces. *Journal of Computational Physics*, 496:112545, 2024. DOI.
9. **Han Zhou**, Jiahe Yang, Wenjun Ying. A kernel-free boundary integral method for the nonlinear Poisson-Boltzmann equation. *Journal of Computational Physics*, 493:112423, 2023. DOI.
10. **Han Zhou**, Wenjun Ying. A Cartesian grid-based boundary integral method for moving interface problems. arXiv:2309.01068.
11. **Han Zhou**, Minsheng Huang, Wenjun Ying. ADI schemes for the heat equation on arbitrary 3D domains and their applications. arXiv:2309.00979.
12. **Han Zhou**, Wenjun Ying. A dimension splitting method for time dependent PDEs on irregular domains. *Journal of Scientific Computing*, 94(1):20, 2023. DOI.
13. Jiacheng Xu, Dan Hu, **Han Zhou**. A phase-field method for elastic mechanics with large deformation. *Journal of Computational Physics*, 471:111630, 2022. DOI.

Talks

- 2026 SIAM Annual Meeting (AN26), Cleveland — Jul. 6–10, 2026
- 2026 CAIMS/SCMAI Annual Meeting, Regina — Jun. 15–19, 2026
- 2026 NCTS Workshop on Mathematics of Living Systems, Hsinchu — May 20–22, 2026
- Kanazawa University - Penn Soft Matter / Applied Math Workshop, Philadelphia — Mar. 23–25, 2026
- Mathematical Modeling, Computational Methods, and Biological Fluid Dynamics: Research and Training, Chicago — Aug. 2025
- Numerical Analysis and PDE Seminar, University of Delaware, Newark, DE — May 2025
- Fluid Mechanics and Waves Seminar, New Jersey Institute of Technology, Newark, NJ — Nov. 2024
- NCTS Seminar on PDE and Machine Learning, Online — Oct. 2024
- The 10th International Congress on Industrial and Applied Mathematics, Tokyo — Aug. 2023
- SICIAM Workshop on Recent Advances in Fast Algorithms, Shenzhen — Aug. 2023
- Chinese Mathematical Society in Computational Mathematics, Nanjing — Jul. 2023
- CSIAM Student Forum, Online — Nov. 2022
- Shanghai Symposium on Scientific and Engineering Computing Methods, Shanghai — Nov. 2021

Honors and Scholarships

- Good Teaching Awards for Spring 2025 (UPenn math) — 2025
- National Scholarship — 2023
- Academic Scholarship for Graduate Students — 2020-2022
- Excellent Bachelor Thesis of School of Mathematical Sciences — 2020

Teaching

University of Pennsylvania

- **Instructor**, AMCS 6035, Numerical and Applied Analysis II, — Spring 2026
- **Instructor**, AMCS 6025, Numerical and Applied Analysis I, — Fall 2025
- **Instructor**, AMCS 6035, Numerical and Applied Analysis II, — Spring 2025
- **Instructor**, AMCS 6025, Numerical and Applied Analysis I, — Fall 2024

Shanghai Jiao Tong University

- **Teaching Assistant** Differential geometry (Fall 2023), Convex optimization (Spring 2022, Fall 2022), Scientific computing (Spring 2021, Fall 2021), Calculus (Fall 2020)

Graduate projects supervision(UPenn):

- Rui Xu (graduate student, AMCS) — 2025
- JiYao Zhang (graduate student, AMCS) — 2025

Technical Skills

Language: Mandarin, English
Computer programming C++, Python, MATLAB.

Academic Service

Organizer, weekly UPenn AMCS Seminar — 2025-2026
Organizer, weekly UPenn Mathematical Biology Seminar — 2024-2025